User Manual

Group 24

1. System Requirements

- Development Environment: Java SE Development Kit 21 or later versions; any compatible development IDE.
- Runtime Environment: JDK 21 or later versions required
- Project language level: 8 (Lambdas, type annotations, etc.)
- Dependencies: Standard JDK 21 only (no third-party libraries)
- Console: Any terminal supporting standard input/output
- Optional: Swing-based GUI is available; CLI must still be supported

2. Get Started

- Launch (CLI only)

Command:

java hk.edu.polyu.comp.comp2021.clevis.Application -html <path>.html -txt <path>.txt

Notes:

– Both -html and -txt are required

– Extensions must be .html and .txt respectively

– On invalid args, you'll see:

Usage error! Correct format: java Application -html log.html -txt log.txt [-gui]

- Launch (CLI only)

Option A: Add -gui flag

java hk.edu.polyu.comp.comp2021.clevis.Application -html <path>.html -txt <path>.txt -gui

Option B: Without -gui, the program prompts:

Do you want to use GUI? (y/n):

Enter y to open GUI; n to stay in CLI.

- Logs

– HTML log: Commands recorded in a table (columns: Index, Command), index starts at 1

– TXT log: One command per line in execution order

– Logs flush on every command; HTML file is properly closed at exit

Example:

Start in CLI mode:

java hk.edu.polyu.comp.comp2021.clevis.Application -html log.html -txt log.txt

Enter the following commands:

rectangle R1 0 0 10 5

circle C1 10 10 5

move R1 5 2
listAll
quit

You should see:

Created rectangle R1
Created circle C1
Moved R1
Circle C1 at (10.00,10.00) with radius 5.00
Rectangle R1 at (5.00,2.00) with width 10.00 height 5.00
Exiting Clevis...

3. Commands Usage

General Rules

- All numeric values are double.
- Shape names are unique.
- All printed numbers are rounded to 2 decimal places.
- All command keywords (i.e., rectangle, circle, list) are case-insensitive
- Argument counts:

– group allows argument count ≥ required (group name + at least one component)
– All other commands require an exact match

3.1 rectangle [n] [x] [y] [w] [h] **Supports Undo/Redo** ✅
Creates a rectangle with top-left corner (x, y), width w, height h. Requires w > 0 and h > 0.
Example:

rectangle R1 0 0 100 50

Output:

Created rectangle R1

Errors:

• The width and height must be greater than 0
• Shape name 'R1' already exists!

3.2 line [n] [x1] [y1] [x2] [y2] **Supports Undo/Redo** ✅
Creates a line segment with endpoints (x1, y1) and (x2, y2).
Example:

line L1 0 0 10 10

Output:

Created line L1

3.3 circle [n] [x] [y] [r] **Supports Undo/Redo** ✅
Creates a circle with center (x, y) and radius r (r > 0).
Example:

circle C1 10 10 5

Output:

Created circle C1

Errors:

• The radius must be greater than 0

• Shape name 'C1' already exists!

3.4 square [n] [x] [y] [l] **Supports Undo/Redo** ✅

Creates a square with top-left corner (x, y) and side length l.

Example:

square S1 3 4 20

Output:

Created square S1

3.5 group [g] [n1] [n2] ... **Supports Undo/Redo** ✅

Groups existing shapes n1, n2, ... into a new shape named g. Group members cannot be directly used by certain operations until ungrouped (see Section 6).

Example:

group G R1 C1

Output:

Created group G

Errors:

• Shape 'R1' not found for grouping (or any missing component)

• Shape name 'G' already exists!

3.6 ungroup [g] **Supports Undo/Redo** ✅

Ungroups shape g back into its component shapes.

Example:

ungroup G

Output:

Ungrouped G

Errors:

• Shape 'G' is not a Group

Note:

• Ungroup will sequentially place the graphics in the group onto the end of the current shapeList, which would give them the highest Z-indicators.

3.7 delete [n] **Supports Undo/Redo** ✅

Deletes shape n. If n is a group, all its members are also deleted.

Example:

delete C1

Output:

Deleted C1

Note:

• If n is a member of a group, direct operations will be restricted.

3.8 boundingbox [n] **Cannot Undo/Redo** ❌
Prints the minimum bounding box of n in “x y w h”.
Example:

boundingbox R1

Output:

0.00 0.00 100.00 50.00

Notes:
- For groups, the bounding box encloses all member shapes.
- Values are printed with 2 decimals.
- If n is a member of a group, direct operations will be restricted.

3.9 move [n] [dx] [dy] **Supports Undo/Redo** ✅
Moves shape n by (dx, dy). If n is a group, all components are moved.
Example:

move R1 5 2

Output:

Moved R1

Note:
- If n is a member of a group, direct operations will be restricted.

3.10 shapeAt [x] [y] **Cannot Undo/Redo** ❌
Returns the name of the topmost shape that covers point (x, y).
Rules:
- Topmost = created later (higher Z-order).
- A point is “covered” if its minimum distance to the outline is < 0.05

Example:

ShapeAt 10 9.97

Outputs:

R1

No shape found at (10.00, 9.97)

Note:
- If n is a member of a group, direct operations will be restricted.

3.11 intersect [n1] [n2] **Cannot Undo/Redo** ❌
Reports whether the bounding boxes of n1 and n2 intersect (share any internal points).
Outputs:

The two shapes intersect

The two shapes do not intersect

Errors:
- Shape 'A' not found / Shape 'B' not found

Note:
- If n is a member of a group, direct operations will be restricted.

3.12 list [n] **Cannot Undo/Redo** ❌
Lists the basic information of shape n.
Example outputs:

Rectangle R1 at (x,y) with width w height h
Square S1 at (x,y) with side l
Circle C1 at (x,y) with radius r
Line L1 from (x1,y1) to (x2,y2)
Group G containing: [A, B, ...]

Errors:
• Shape 'n' not found
Note:
• If n is a member of a group, direct operations will be restricted.

3.13 listAll **Cannot Undo/Redo** ❌
Prints all shapes from top to bottom (reverse creation order). For groups, prints nested contents with two-space indentation per level.
Example:

Group G containing: [R1, C1]
  Rectangle R1 at (0.00,0.00) with width 100.00 height 50.00
  Circle C1 at (10.00,10.00) with radius 5.00

Note:
• If there is no any shape, the output will be empty.

3.14 undo **Cannot Undo/Redo** ❌
Reverses the most recent undoable operation and pushes it to the redo stack.
Output:

Undo: <Command Name>
Nothing to undo (if none)

3.15 redo **Cannot Undo/Redo** ❌
Reapplies the most recently undone operation and pushes it back to the undo stack.
Output:

Redo: <Command Name>
Nothing to redo (if none)

3.16 quit **Cannot Undo/Redo** ❌
Quits Clevis and ends the process.
Output:

Exiting Clevis...

Grouping Rules
• After grouping, members cannot be directly used by the following until ungrouped:
– delete, move, boundingbox, list, intersect
• If a shape belongs to a group, ungroup the parent first:

– ungroup [groupName], then operate on members
• Attempts to operate directly on a grouped member will be rejected by validation.

GUI Guide (Optional Bonus)
• Launch:
– Use -gui flag, or answer y when prompted: "Do you want to use GUI? (y/n):"
• Layout:
– Left: scrollable drawing panel
– Right: command history (monospaced)
– Bottom: command field, Execute button, Zoom In/Out
• Behavior:
– Commands are processed by the same model as CLI
– Logs are written to the same HTML/TXT files
– Zoom controls scale the view; initial scale fits overall bounding box
– Enter quit to close and exit
• Startup notes:
– Provide valid .html and .txt paths (same as CLI)
– GUI will load and display existing logs

4.Troubleshoot

Q: I get a usage error at startup.
A: Ensure both -html and -txt are provided with correct extensions:
java hk.edu.polyu.comp.comp2021.clevis.Application -html log.html -txt log.txt [-gui]

Q: The command is not recognized.
A: You will see "Error: Unknown command 'xxx'". Check spelling and argument counts.

Q: Wrong number of arguments?
A: You will see "Error: Invalid number of arguments for command 'cmd'".
Note group requires: group g n1 n2 ... (at least one member).

Q: Name already exists or not found?
A: On creation/group name clashes: "Shape name 'n' already exists!" When querying a missing shape: "Shape 'n' not found" (or a similar message per command)

Q: Why can't I operate on a shape that's inside a group?
A: delete, move, boundingbox, list, intersect are restricted on group members. Ungroup first.

Q: Undo/redo does nothing.
A: You'll see "Nothing to undo" or "Nothing to redo" when the stacks are empty.

Q: GUI won't start or fails to log.

A:
• GUI startup failed: <reason>
• Error initializing logger: <reason>
• Error loading log file: <reason>
Ensure valid .html/.txt paths and write permission.